

Computational Efficient DSVM-Based Model Predictive Direct Speed Control for SPMSM Drives With Mechanical Disturbance Suppression

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Abstract—This article proposes a simplified discrete space vector modulation (DSVM)-based model predictive direct speed control (MPDSC) with an improved load disturbance observer for permanent magnet synchronous motor (PMSM) drives. First, a simplified DSVM method is used to improve the steady-state performance of MPDSC. In this DSVM method, a novel geometric method relying only on three auxiliary lines in each sector is designed to simplify the algorithm's complexity. In this way, the set of candidate vectors is quickly determined. Then, the current pulsation and speed of MPDSC are suppressed, and the computational burden of the DSVM execution process is reduced. Second, the reasons that affect the dynamic performance of the conventional linear extended state observer (ESO)-based mechanical disturbance observer are analyzed, and the observed error of the observer is derived. Based on the observer error, an improved mechanical disturbance observer is proposed to accelerate the convergence process. The Lyapunov theory proves the stability of the proposed observer. Finally, the feasibility of the proposed method is verified by experiments.

Index Terms—Discrete space vector modulation (DSVM), mechanical disturbance observer, model predictive control (MPC), permanent magnet synchronous machines.

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs) are widely used in electric vehicles, CNC machine tools, industrial robots, aerospace, and other applications because of their simple structure and high power density [1]. Model predictive control (MPC) strategies are a widely used method in PMSM drives, primarily due to their straightforward

integration of multicontrol variable optimization, robust capacity to address nonlinearities, and notable dynamic performance [2], [3]. According to different control objectives, MPC strategies in the PMSM drive system can be categorized into model predictive current control (MPCC) [4], model predictive torque control (MPTC) [5], and so on.

However, the conventional MPC methods require a linear proportional–integral (PI) speed controller to generate the current or torque reference in a cascade fashion. This structure can improve the performance of the system to some extent. However, its dynamic performance is still limited [6]. The dynamic performance of the system is limited due to the constrained bandwidth of the outer loop PI speed controller. However, in some applications such as industrial traction, steel mill drive, and annealing lines, the transient speed performance is crucial [7]. To inherit the excellent dynamic performance of MPC and further accelerate the speed loop control, model predictive direct speed control (MPDSC) has received much attention in recent years. Different from other MPC methods, MPDSC uses one controller that integrates both motor speed and current, which simplifies the control system structure [8].

Nevertheless, the conventional MPDSC method only evaluates the responses under the action of basic voltage vectors (VV) and selects the optimal VV through a cost function [8]. In this way, the PMSM drive system suffered relatively large current and speed fluctuations, resulting in poor steady-state performance [9]. To improve the steady-state performance of the MPDSC, [10] proposes a duty ratio optimization-based MPDSC with a new dual cost function for PMSM drives to improve the steady-state performance. In this method, the duty cycle of the output vector is derived according to the deadbeat speed control principle. Then the amplitudes of the basic VVs are optimized by zero VVs in each control period. Furthermore, a double-vector-based MPDSC with a no weighting factor cost function is proposed in [11]. Although the multivector MPC methods can improve the steady-state performance of MPDSC to some extent, the duty cycle calculation will increase the algorithm complexity.

In recent years, discrete space vector modulation (DSVM) has attracted more and more attention [12]. The DSVM strategy synthesizes a large number of virtual VVs (VVs) by dividing the control period into equal intervals. In this way, the steady-state performance can be improved by increasing the elements of the VV set. Then, the action times of multiple

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vectors within a control period are fixed, which is very intuitive and simple to implement [13]. However, since the number of VVVs in the extended vector set increases quadratically with the increase in time intervals, the difficulty of selecting the optimal VV also increases. To reduce the number of candidate VVs, more attention has been focused on the issue. In [14], considering the position of the flux linkage vector, the author reduced the elements of the predetermined lookup table (LUT) by the error of the torque, flux linkage, and their corresponding reference values. The advantage of this method is that the reduced VV control set is determined in a DTC-like manner without any hysteresis comparators. In [15], 30 VVVs based on DSVM for induction motor drive are synthesized, and the optimal sector is obtained by evaluating six central VVVs at the first stage, which reduces the number of candidate voltages. Furthermore, the duty cycle of the optimal VV is obtained in the mathematical solution, which solves the partial derivative equation between the VVVs [16]. Gu et al. [17] propose an efficient DSVM-based MPTC with constant switching frequency, where the torque ripples and current harmonics are reduced. This method determines the optimal set of VVs in three steps; however, its implementation logic is complex. However, designing the LUT increases the complexity of the algorithm implementation. Similarly, the DSVM method in [18] requires the determination of the categories by the cost function and then traversing through the cost function to obtain the final output vector. In addition, the method needs to obtain the final output vector by traversing through the cost function after determining the categories. This process not only requires the design of multiple LUTs but also requires the cost function to determine the categories and output vectors, which increases the complexity of algorithm implementation. Wang et al. [19] propose a novel DSVM method in a deadbeat torque flux control solution, which effectively reduces the number of VV candidates to three. However, the calculation process to find the three candidate VVs is complicated. After obtaining the optimal VV from the three candidate VVs, the selected optimal VV is difficult to convert into switching sequences.

In addition, in actual traction applications, PMSM systems are always confronted with different mechanical disturbances, which include internal disturbances, such as friction force and unmodeled dynamics, and external disturbances, such as load disturbances [20]. To reduce the influence of mechanical disturbance, parameter identification methods [21] and disturbance observers [22] are commonly used in motor drive systems. Observer-based methods are more widely used to improve the disturbance rejection due to their relatively simple design and implementation. Generally, the observers include the Luenberger observer [23], the Kalman filter observer [24], the sliding mode observers (SMOs) [25], the extended state observer (ESO) [26], etc. SMO-based disturbance observers can achieve a fast estimation effect but often suffer from large estimation fluctuations. Among different disturbance observers, ESO-based disturbance observers are widely used due to their decisive influence on the disturbance rejection performance of the system [27]. Moreover, the parameter setting of the linear ESO (LESO) is convenient, and the stability analysis is easy, so it is a popular choice [25]. However, the estimated

total mechanical disturbance of the conventional second-order LESO-based load observer is designed based on the estimation error of the speed rather than the estimation error of the total mechanical disturbance. In this case, convergence of the speed estimation is a prerequisite for convergence of the observer, and then the estimation error of the speed is too small. At this time, it is difficult to more rapidly estimate total mechanical disturbance with a small gain factor [28], [29]. Nakao et al. [30] present a mechanical disturbance observer incorporating first-order lag filter to suppress high-frequency noise in a load disturbance observer. However this observer does not observe the speed and does not consider the effect of speed disturbance on the controller and load observation.

This article proposes a simplified DSVM-based MPDSC with an improved load disturbance observer to improve control performance and mechanical disturbance rejection ability. The main contributions are listed below.

- 1) First, the deadbeat control principle is used to establish a link between the mechanical model and the current model of PMSM. Based on q -axis reference current i_q^* , which is obtained from the mechanical model, the reference VV (RVV) can be derived to combine the MPDSC method and DSVM technology in this article. Some experiments have shown the feasibility and better tracking performance of this method.
- 2) Aiming to simplify complexity and reduce the computational burden of DSVM, a novel optimal VV selection mode based on three auxiliary lines is proposed. The area where the optimal VV is resided is first narrowed to a certain sector. Within the sector, the position where the optimal subregion is located is quickly determined by the distance from the RVV to the three auxiliary lines. Then, the execution process of the DSVM method is optimized, and the execution simplicity of the proposed DSVM method is experimentally verified.
- 3) In addition, to suppress the mechanical interference effect of the proposed MPDSC, a load observer based on the conventional ESO framework is proposed. In this observer, the second-order observation characteristics are reduced to first-order to increase the observation bandwidth by adding the derivative of the speed observation error derivative to the load observation. In this way, under the same parameters of the conventional ESO-based load observer, the proposed load observer has a better dynamic performance, which can satisfy the quick requirement of the MPDSC.

The rest of this article is organized as follows. In Section II, the mathematical model of the SPMSM system is introduced. Then, in Section III, the proposed DSVM-based MPDSC is described in detail. In Section IV, the conventional LESO-based load disturbance observer is analyzed, and an improved load disturbance observer is introduced. The experimental setup and results are given in Section V. Finally, Section VI concludes this article.

II. MATHEMATICAL MODEL OF SPMSM AND DSVM PRINCIPLE

A. Mathematical Model of SPMSM

Under the assumptions that the magnetomotive force is sinusoidal across the air gap, the winding of the PMSM is

symmetrical, and core loss and eddy current can be neglected, the mathematical current model of the PMSM in the dq -axis rotating reference frame is expressed as follows [6]:

$$\begin{cases} u_d = L_s \frac{di_d}{dt} + R_s i_d - \omega_e L_s i_q \\ u_q = L_s \frac{di_q}{dt} + R_s i_q + \omega_e (L_s i_d + \psi_f) \end{cases} \quad (1)$$

$$T_e = \frac{3}{2} n_p \psi_f i_q \quad (2)$$

where L_s and R_s represent the stator resistance and the stator inductance, respectively; i_d and i_q are the dq -axis currents; ψ_d and ψ_q are the dq -axis stator flux; ω_e is the electrical rotor speed (in rad/s); and ψ_f is the flux linkage. u_d and u_q represent the dq -axis voltages.

Moreover, the mechanical equation can be expressed as

$$\frac{d\omega_m}{dt} = \frac{1}{J} (T_e - T_L) - \frac{1}{J} B \omega_m \quad (3)$$

where ω_m is the mechanical rotor speed, T_L is the load torque, J is the rotational inertia, and B is the viscous friction coefficient.

The mechanical time constant is significantly greater than the electrical time constant, suggesting that the response time of the current loop is faster than that of the speed loop. Consequently, differing time intervals of discretization exist between the current model equations, encompassing (1) and (2), and the mechanical model equation, as described in (3). In the design of experimental platforms, the time interval for the mechanical model equation typically ranges from 4 to 20 times that of the current model equation [11]. In this study, we adopt a multiple of 10, denoted as $c = 10$, in the subsequent equations.

Then, from (1) to (3), the SPMSM discrete model is deduced as

$$\begin{cases} i_d^{k+1} = i_d^k - \frac{R_s T_s}{L_s} i_d^k + T_s \omega_e^k i_q^k + \frac{T_s}{L_s} u_d^k \\ i_q^{k+1} = i_q^k - \frac{R_s T_s}{L_s} i_q^k - T_s \omega_e^k i_d^k + \frac{T_s}{L_s} u_q^k - \frac{\psi_f T_s \omega_e^k}{L_s} \end{cases} \quad (4)$$

$$\omega_m^{ck+c} = \frac{cT_s}{J} (T_e^{ck} - T_L) + \left(1 - \frac{cT_s B}{J}\right) \omega_m^{ck} \quad (5)$$

$$T_e^{k+1} = \frac{3}{2} n_p \psi_f i_q^{k+1} \quad (6)$$

where T_s is the current discrete time interval, $i_{d,q}^k$ and $i_{d,q}^{k+1}$ are the dq -axis stator currents at k and $k+1$ instant, respectively, and $u_{d,q}^k$ are the d -axis and q -axis stator voltages at k instant. T_e^{k+1} is the electrical torque at $k+1$ instant. ω_m^k and ω_m^{ck+c} are the mechanical speed at instant k and instant $ck+c$, respectively.

B. VV Synthesis by DSVM Principle

According to the principle of DSVM, in each control period, extended VVVs can be synthesized by applying several VVs for equal time intervals [16]. For a three-phase two-level voltage-source inverter, if a sampling period is divided equally into N parts, the VVs in the extended control set synthesized by DSVM can be expressed as follows:

$$v_{\text{DSVM}} = \sum_{i=1,2,\dots,N} t_i V_i^{\text{basic}} \quad (7)$$

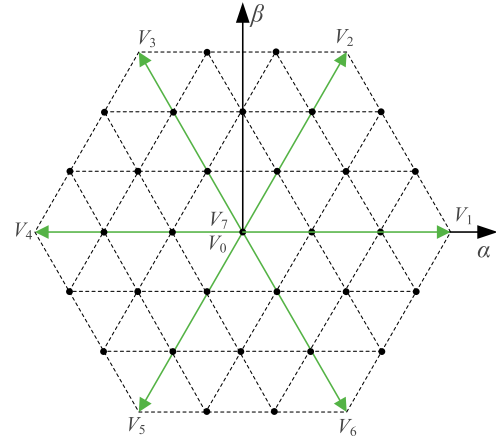


Fig. 1. Extended VV control set by DSVM principle.

where $V_i^{\text{basic}} = V_{0,\dots,7}$ from the three-phase two-level voltage-source inverter, $t_i = T_s/N$, and N is the interval number of each sampling period.

Then, the total VVs' number of the extended control set, n_{total} , can be deduced as

$$n_{\text{total}} = 3N^2 + 3N + 1. \quad (8)$$

Specifically, for compromising the switching frequency and steady-state performance, only the case of $N = 3$ is discussed. The number of total VVs at this point is 38. When $n = 3$, the VVs in extended control set can be shown as in Fig. 1. The green VVs in Fig. 1 are the basic VVs of the three-phase two-level voltage source inverter, while the other black VVs are the synthesized vectors according to the DSVM principle.

III. IMPROVED DSVM-BASED MPDSC METHOD

A. RVV Acquisition

The conventional MPDSC only evaluates the eight basic VVs by a cost function, and it is not easy to obtain a more desirable output voltage. Undesirable VV output can lead to poor steady-state performance for both current and speed. Thus, the DSVM method is introduced into the MPDSC to improve the steady-state performance in this article. However, conventional DSVM methods often need to traverse a large number of VVVs by cost functions or offline LUTs, so an improved voltage selection method is proposed in this section to simplify the selection process. According to the deadbeat control principle, $i_d^{k+1} = i_d^* = 0$ and $i_q^{k+1} = i_q^*$ are considered in the derivation process, then the reference VV can be obtained by

$$\begin{cases} u_d^* = R_s i_d^k - L_s \omega_e^{ck} i_q^k + \frac{L_s}{T_s} (0 - i_d^k) \\ u_q^* = R_s i_q^k + L_s \omega_e^{ck} i_d^k + \frac{L_s}{T_s} (i_q^* - i_q^k) + \psi_f \omega_e^{ck} \end{cases} \quad (9)$$

where the superscript “*” denotes the reference value. Since there is no q -axis current reference i_q^* in MPDSC, i_q^* is often obtained by the speed prediction equation [15]. The mechanical time constant is much larger than the electrical time constant, which means the q -axis current between cT_s and $(c+1)T_s$ changes many times. To simplify the derivation

process, $i_q^{ck} = i_q^{k+1}$ is set in the mechanical discrete model. Then, according to (5) and (6), the mechanical discrete model can be obtained by

$$\omega_m^{ck+c} = \frac{cT_s}{J} \left(\frac{3}{2} n_p \psi_f i_q^{k+1} - T_L \right) + \left(1 - \frac{cT_s B}{J} \right) \omega_m^{ck}. \quad (10)$$

Meanwhile, based on the deadbeat control principle, the q -axis current reference i_q^* can be expressed as

$$i_q^* = \frac{2}{3n_p \psi_f} \left[\frac{J}{cT_s} (\omega_m^* - \omega_m^{ck}) + B\omega_m^{ck} + T_L \right]. \quad (11)$$

Considering that both the power converter and the motor have the maximum allowable current in the actual PMSM drive system, it is necessary to limit the reference q -axis current. In this article, the q -axis current is limited to 6 A. Then, the reference VV can be derived from (9) and (11)

$$\begin{cases} u_d^* = R_s i_d^k - L_s \omega_e^{ck} i_q^k + \frac{L_s}{T_s} (0 - i_d^k) \\ u_q^* = R_s i_q^k + \psi_f \omega_e^{ck} + L_s \omega_e^{ck} i_d^k + \frac{L_s}{T_s} \left\{ \frac{2}{3n_p \psi_f} \left[\frac{J}{cT_s} \omega_m^* - \left(B - \frac{J}{cT_s} \right) \omega_m^{ck} + T_L \right] - i_q^k \right\} \end{cases} \quad (12)$$

The RVV can be transformed into the $\alpha\beta$ -axis by the following equation:

$$\begin{cases} u_\alpha^* = u_d^* \cos \theta - u_q^* \sin \theta \\ u_\beta^* = u_d^* \sin \theta + u_q^* \cos \theta \end{cases} \quad (13)$$

where θ is the electric rotor angle, and u_α^* and u_β^* are the α β -axis components of RVV. Then, the information of the RVV, including the reference angle θ_{ref} and the amplitude $|u_{\text{ref}}|$ of the RVV, can be given by

$$\theta_{\text{ref}} = \arctan \left(\frac{u_\beta^*}{u_\alpha^*} \right) \quad (14)$$

$$|u_{\text{ref}}| = \sqrt{(u_\alpha^*)^2 + (u_\beta^*)^2}. \quad (15)$$

B. Optimal Subregion Selection Mode

Increasing the number of VVs in the control set can enhance the precision of output VV, which will improve the steady-state performance of speed [16]. However, evaluating all 38 candidate VVs within a short control period imposes a significant computational burden, especially in practical motor control systems using microprocessors. To address the above problems, this section proposes a DSVM selection mode with a geometrical approach to simplifying the optimal VV determination process.

However, DSVM-based control theory divides the whole range of the hexagon modulation region into many equal small triangular regions. But in most cases, the VV used to provide optimal performance over short intervals lies only in a narrower range. This means that most sectors are not necessarily considered in the original mode of subdividing regions. To reduce the elements of the VV control set, only the sector in which the RVV is located is divided by the DSVM principle. It is worth noting that the basic VV in front of

TABLE I
BASIC VECTORS IN EACH SECTOR

Sector	V_{opt1}	V_{opt2}
Sector I	V_4	V_6
Sector II	V_6	V_2
Sector III	V_2	V_3
Sector IV	V_3	V_1
Sector V	V_1	V_5
Sector VI	V_5	V_4

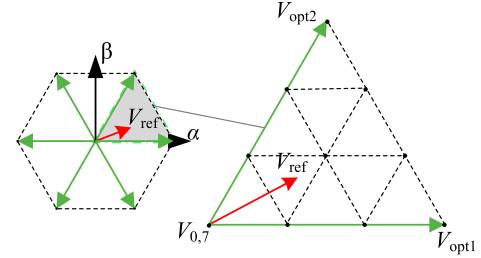


Fig. 2. Sector subdivision based on DSVM.

the RVV counterclockwise rotation is marked as V_{opt1} , and vice versa is V_{opt2} . These two specific vectors will change due to different sectors. In this way, the sector sequence and the corresponding two operating vectors can be obtained if the certain sector is selected as Table I. As shown in Fig. 2, any extended VVs in a certain sector can be composed of two operating vectors V_{opt1} and V_{opt2} and zero vector. Then, the extended VV control set of the sector where RVV resides can be expressed as

$$v_{\text{DSVM}} = \sum_{j=1,2,\dots,N} t_j V_{\text{optj}} \quad (16)$$

where $V_{\text{optj}} = V_{\text{opt1}}, V_{\text{opt2}}$ or $V_{0,7}$.

Even though the above procedures can narrow the range of feasible subregions down, ten VVs still need to be evaluated. To further narrow down the selection range of extended control set and to simplify the complexity of the algorithm, a method based on three auxiliary lines is proposed in detail to select the optimal subregion directly.

From Table I, two operating vectors in any sector can be obtained. Then, as shown in Fig. 3, take the zero vector as the origin of coordinates, the direction of V_{opt1} as the x -axis, and 90° ahead of the direction of V_{opt1} as the y -axis to make a Cartesian coordinate system. The operating vectors will change with the sector change. At the same time, auxiliary lines l_1 , l_2 , and l_3 take the direction of V_{opt1} , the direction of V_{opt2} , and the direction of the angle bisector of angle $V_{\text{opt1}} - V_{0,7} - V_{\text{opt1}}$, respectively. For calculating the mentioned distances, from (14), the residual angle θ_δ of the reference angle at the sector where the RVV resides can be obtained as

$$\theta_\delta = \text{Mod} \left(\frac{\theta_{\text{ref}}}{\pi/3} \right) \quad (17)$$

where Mod is the remainder function.

The distances from the RVV to the auxiliary lines l_1 and l_2 are denoted as d_1 and d_2 , respectively. In particular, the projection of the RVV in l_3 directions is denoted as d_3 . Their

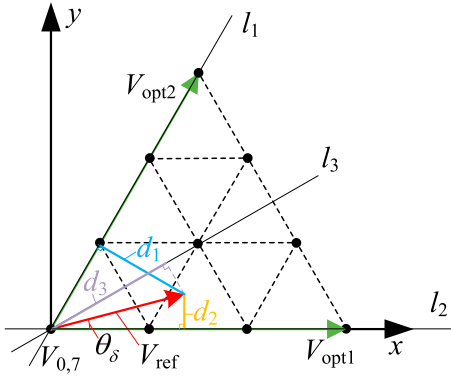


Fig. 3. Schematic of optimal subregion selection mode.

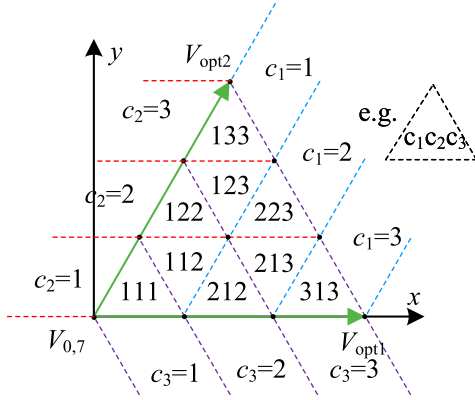


Fig. 4. All cases under different RVVs in certain sectors.

specific expressions are described in the following equations:

$$\begin{cases} d_1 = |u_{ref}| \sin \theta_\delta \\ d_2 = |u_{ref}| \sin \left(\frac{\pi}{3} - \theta_\delta \right) \\ d_3 = |u_{ref}| \cos \left(\frac{\pi}{6} - \theta_\delta \right) \end{cases} \quad (18)$$

where $|u_{ref}|$ can be obtained from (15).

The above three distances are equivalent to the components in the three directions of l_1 , l_2 , and l_3 ; they can represent the size of V_{opt1} , V_{opt2} , and V_0 , respectively. Then, d_1 , d_2 , and d_3 are rounded up. The number of layers c_1 , c_2 , and c_3 in the three directions of d_1 , d_2 , and d_3 can be given as

$$\begin{cases} c_1 = \text{ceil} \left(\frac{3\sqrt{3}d_1}{|u_{dc}|} \right) \\ c_2 = \text{ceil} \left(\frac{3\sqrt{3}d_2}{|u_{dc}|} \right) \\ c_3 = \text{ceil} \left(\frac{3\sqrt{3}d_3}{|u_{dc}|} \right) \end{cases} \quad (19)$$

where “ceil” is the up-integer function.

According to the values of c_1 , c_2 , and c_3 , all cases of RVV in any triangular subregions can also be represented in Fig. 4. Similarly, no matter which sector the RVV is in, all the following cases are the same. For example, subregion 313 represents c_1 layer is 3, c_2 layer is 1, and c_3 layer is 3. The optimal subregion is obtained directly according to the

distance between the RVV and the three auxiliary lines. The specific triangular subregion where the RVV resides can be directly determined by the number of layers c_1 , c_2 , and c_3 .

Then, depending on different orientations of the triangle subregions, the expressions for the candidate VVs $V_{can1,2,3}$ on the three vertices can be expressed as

$$\begin{cases} V_{can1} = \frac{c_1}{3} V_{opt1} + \frac{c_2 - 1}{3} V_{opt2} \\ V_{can2} = \frac{c_1 - 1}{3} V_{opt1} + \frac{c_2}{3} V_{opt2} \\ V_{can3} = \frac{c_1 - 1}{3} V_{opt1} + \frac{c_2 - 1}{3} V_{opt2}, \\ \quad \text{if } (c_1 + c_2 + c_3) \text{ is odd} \\ V_{can3} = \frac{c_1}{3} V_{opt1} + \frac{c_2}{3} V_{opt2}, \quad \text{if } (c_1 + c_2 + c_3) \text{ is even} \end{cases} \quad (20)$$

where $V_{can1,2,3}$ are three candidate VVs. V_{opt1} and V_{opt2} can be obtained from Table I.

From the above procedures, the number of layers can be used to determine the specific triangular subregion where the RVV resided in. Then, this method can directly determine the three candidate VVs $V_{can1,2,3}$ through equation (20). This process can further narrow down the ten VVs in the DSVM control set to 3 using geometric principles, which reducing the algorithmic complexity and computational burden.

C. Cost Function

The cost function for the VV is used in this article, which is given as

$$g = |u_d^* - u_{d-i}| + |u_q^* - u_{q-i}| \quad (21)$$

where u_{d-i} and u_{q-i} are the d -axis and q -axis voltages of the three VV candidates from (20), respectively. Obviously, the cost function introduced in (21) avoids the excessive tuning works of the weighting factor in MPDSC. Then, the candidate VVs at the vertexes of the optimal triangular subregion are evaluated by the cost function, and the optimal VV is determined.

IV. IMPROVED LOAD DISTURBANCE OBSERVER

Since the MPDSC is a noncascade control structure that directly integrates the mechanical and current models through the cost function, mechanical disturbance can directly affect the MPDSC method by introducing errors in RVV acquisition. From this perspective, it is necessary to suppress the perturbation by taking the measures in this article to suppress the load through observation.

A. Analysis of Conventional ESO-Based Mechanical Disturbance Observer

The load observer based on the conventional LESO can be designed according to the mechanical model (3). Here, f_d is the total mechanical disturbance, including the disturbance caused by the change in mechanical equation parameters, such as J , ψ_f , and B , the load change, and the external disturbance.

The dynamic motion equation of the SPMSM that considers machine parameters and external disturbances is expressed as

$$\begin{aligned} \frac{d\omega_m}{dt} &= \frac{3n_p\psi_{f0}}{2J_0}i_q - \frac{B_0\omega_m}{J_0} + f_d \\ &= a_0i_q - b_0\omega_m + f_d \end{aligned} \quad (22)$$

where $a_0 = 3n_p\psi_{f0}/2J_0$, $b_0 = B_0/J_0$, and the subscript “0” denotes nominal parameter values.

Subtract (3) from (22), the total mechanical disturbance f_d can be expressed as

$$f_d = -\frac{\Delta J}{J} \frac{d\omega_m}{dt} - \frac{\Delta B\omega_m}{J_0} + \frac{3n_p\Delta\psi_{f0}}{2J_0}i_q - \frac{T_L}{J_0} + f_e \quad (23)$$

where J , B and ψ_f are the actual parameter values, which will change under different conditions, $\Delta J = J - J_0$, $\Delta B = B - B_0$ are the errors of actual parameters, and f_e is the disturbance including the load change and the external disturbances.

According to (22), the conventional LESO-based mechanical disturbance observer can be designed as

$$\begin{cases} e_1(t) = z_1(t) - \omega_m(t) \\ \dot{z}_1(t) = z_2(t) + a_0i_q(t) - b_0\omega_m(t) - \beta_1e_1(t) \\ \dot{z}_2(t) = -\beta_2e_1(t) \end{cases} \quad (24)$$

where $z_1 = \hat{\omega}_m$ and $z_2 = \hat{f}_d$ are the estimated values of mechanical speed and the total mechanical disturbance, respectively. β_1 and β_2 are the gains of ESO, which can be designed according to [31]. In addition, when β_1 and β_2 are strictly positive, the Hurwitz stability criterion is satisfied, and estimation errors can converge exponentially to zero.

According to (24), the mechanical equation (22) can be rewritten as

$$\begin{cases} \dot{\omega}_m = a_0i_q(t) - b_0\omega_m(t) + f_d \\ \dot{f}_d = \xi(t) \end{cases} \quad (25)$$

where $\xi(t)$ is the derivation of f_d .

From (24) and (25), the estimated error of z_1 and z_2 can be obtained as

$$\begin{cases} \dot{e}_1(t) = z_2(t) - \dot{\omega}_m(t) - \beta_1e_1(t) \\ \dot{e}_2(t) = \xi(t) - \beta_2e_1(t) \end{cases} \quad (26)$$

where $\dot{e}_1(t)$ and $\dot{e}_2(t)$ are the derivatives of $e_1(t)$ and $e_2(t)$, respectively.

From (26), the dynamic estimate error of z_2 can be deduced as

$$e_2(t) = z_2(t) - x_2(t) = \dot{e}_1(t) + \beta_1e_1(t). \quad (27)$$

From the above equation, it is clear that the observation error of e_2 consists not only of e_1 but also of the first-order derivative $\dot{e}_1$ of e_1 .

Under the above coefficient setting conditions, the estimated values z_1 and z_2 of the observer will gradually converge to the real x_1 and x_2 , so as to achieve the purpose of real-time observation. In the convergence process of the observer, the convergence of z_1 is the most important because the observer in (24) is based on the error e_1 of z_1 and x_1 . Thus, the convergence of the estimated speed z_1 to the actual speed x_1 is a prerequisite for the convergence of the observer. However, the total mechanical disturbance z_2 can only converge slowly

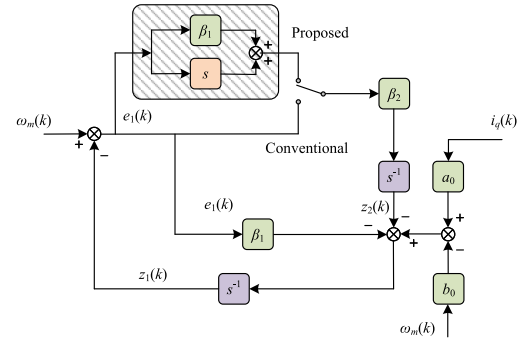


Fig. 5. Improved LESO-based load disturbance observer.

after z_1 converged according to the error e_1 about z_1 and x_1 rather than the error of z_2 . In fact, in the load observer in (24), only z_2 with slow convergence is used to estimate the actual load, which is obviously not reasonable.

B. Improved Mechanical Disturbance Observer

From the analysis in Section IV-A, it is necessary to design an LESO-based load observer that accelerates the convergence of z_2 to achieve faster estimation speeds to cater to MPDSC's rapidity. In this section, an improved LESO-based load disturbance observer is designed by analyzing the observation error of z_2 .

Then, according to the dynamic estimate error discussed in (27), the improved LESO-based load disturbance observer is designed as follows:

$$\begin{cases} e_1(t) = z_1(t) - \omega_m(t) \\ \dot{z}_1(t) = z_2(t) + a_0i_q(t) - b_0\omega_m(t) - \beta_1e_1(t) \\ \dot{z}_2(t) = -\beta_2[\dot{e}_1(t) + \beta_1e_1(t)] \end{cases} \quad (28)$$

Fig. 5 shows the control block diagram of the improved LESO for mechanical disturbances based on (28).

For ease of analysis, f_d is defined as a non-time-varying constant. From (26) and (28), define estimate error $r_1(t) = e_1(t)$, $r_2(t) = \dot{r}_1(t) = e_2(t) - \beta_1e_1(t)$, the system state-space equation of the observer error is

$$\begin{bmatrix} \dot{r}_1(t) \\ \dot{r}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\beta_1\beta_2 & -(\beta_1 + \beta_2) \end{bmatrix} \begin{bmatrix} r_1(t) \\ r_2(t) \end{bmatrix} - \begin{bmatrix} 0 \\ \xi(t) \end{bmatrix}. \quad (29)$$

For the stability proof, demonstrate that error equilibrium points of ILESO are asymptotically stable over a wide range at the steady-state stage. Then, the following equation holds at the steady-state stage:

$$\begin{bmatrix} \dot{r}_1(t) \\ \dot{r}_2(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (30)$$

Define the first-order derivative of f_d of the above observer, $\xi(t)$, is bounded, then the following condition holds:

$$|\xi(t)| \leq \xi_0 \quad (31)$$

where ξ_0 is a positive constant.

Then, from (28), (29) and the condition (30), the quantitative observation error of the ILESO is

$$e_1(t) \leq \frac{\xi_0}{\beta_1\beta_2}, \quad e_2(t) \leq \frac{\xi_0}{\beta_2}. \quad (32)$$

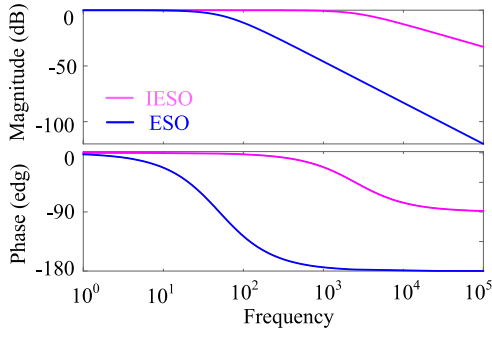


Fig. 6. Bode diagram of the conventional ESO-based load disturbance observer and the improved one.

Similarly, according to (24), define $r_1(t) = e_1(t) = z_1(t) - x_1(t)$ and $\dot{r}_1(t) = r_2(t) = -\beta_1 e_1(t) + e_2(t)$, the observation error state-space equation of the conventional ESO-based load observer is

$$\begin{bmatrix} \dot{r}_1(t) \\ \dot{r}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\beta_2 & -\beta_1 \end{bmatrix} \begin{bmatrix} r_1(t) \\ r_2(t) \end{bmatrix} - \begin{bmatrix} 0 \\ \xi(t) \end{bmatrix}. \quad (33)$$

And the quantitative observation error of the conventional LESO is

$$e_1(t) \leq \frac{\xi_0}{\beta_2}, \quad e_2(t) \leq \frac{\beta_1 \xi_0}{\beta_2}. \quad (34)$$

Since β_1 is usually greater than 1 at the practical parameter settings, it is clear from the quantitative observation errors of the two observers in (32) and (34) that ILESO has a better observation performance.

Based on the improved observer (28), the estimation error e_1 of z_1 is no longer used by the estimation equation of z_2 . In contrast, the observation error of z_2 that contains the first-order derivative of e_1 is adopted to accelerate the convergence process of the estimation equation of z_2 . Therefore, excellent dynamic performance can be obtained.

To analyze the performance of the designed load observer, according to (22) and (24), eliminating x_1 and z_1 , and performing the Laplace transform on the differential, the transfer function from z_2 to x_2 of the conventional ESO-based load disturbance observer is obtained as

$$\frac{z_2}{x_2} = \frac{\beta_2}{s^2 + \beta_1 s + \beta_2}. \quad (35)$$

Correspondingly, for the IESO proposed in this article, the transfer function of the IESO can be obtained according to (28) and (22) and expressed as

$$\frac{z_2}{x_2} = \frac{\beta_2}{s + \beta_2}. \quad (36)$$

The gain coefficient ω_0 of the two observers is set to the same 50, at which point the corresponding Bode diagram of the two observers can be shown in Fig. 6.

It can be obtained that the improved ESO introduces the observation error of e_2 into z_2 to obtain a larger bandwidth without changing the basic structure of the conventional ESO-based load observer. It can be seen from the Bode diagram that the proposed load observer maintains a relatively stable gain over a wide frequency range, which not only

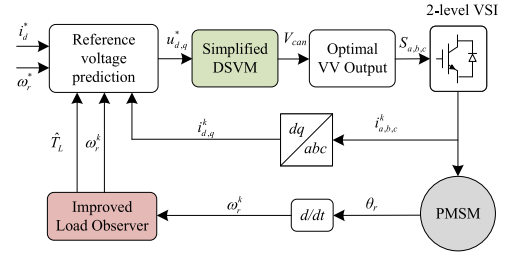


Fig. 7. Block diagram of the proposed MPDSC method.

improves the steady-state performance but also accelerates the estimated response of z_2 . Correspondingly, the conventional ESO-based load disturbance observer has a negative gain at low frequencies, which makes errors when estimating load disturbances. Moreover, the phase is shifted, so the load estimation is delayed. So, the load observer proposed in this article can have a faster response, which is more catering to the fast characteristics of MPDSC.

Next, the Lyapunov theory is used to prove the stability of the proposed load observer. Then, the constructed Lyapunov function V is chosen as

$$V = \frac{1}{2} \beta_1^2 \beta_2^2 r_1^2(t) + \frac{1}{2} \beta_1 \beta_2 r_2^2(t). \quad (37)$$

The derivation of the constructed Lyapunov function $\dot{V}$ is

$$\dot{V} = \beta_1^2 \beta_2^2 r_1(t) \dot{r}_1(t) + \beta_1 \beta_2 r_2(t) \dot{r}_2(t). \quad (38)$$

From (29), (38) can be rewritten as

$$\dot{V} = -(\beta_1^2 \beta_2 + \beta_1 \beta_2^2) r_2^2(t). \quad (39)$$

From Section IV-A, the coefficients β_1 and β_2 are strictly positive, then the derivative of V is less than zero constantly, then V is positively determined [32]. Then, the asymptotical stability of the designed LESO is proved. It is worth noting that the parameter design of (28) can still be based on [31].

Then, the equivalent load, including the total mechanical disturbance, is estimated as

$$\hat{T}_L = -\frac{f_d}{J_0} \quad (40)$$

where $\hat{T}_L$ is the observed load which can be obtained by z_2 . According to (11) and (35), the reference value of the compensated q -axis current can be rewritten as

$$i_q^* = \frac{2}{3n_p \psi_f J_0} \left[\frac{J_0}{c T_s} (\omega_m^* - \omega_m^{ck}) + B_0 \omega_m^{ck} + \hat{T}_L \right]. \quad (41)$$

Fig. 7 shows the block diagram of the proposed DSVM-based MPDSC with improved load disturbance observer for PMSM drives. Since the DSVM method proposed in this article determines the optimal vector selection using only the reference voltage, the method can be generalized to any other ac motor (including IPMSM, IM, etc.) drive that has access to obtain the reference voltage. And the proposed mechanical perturbation observer can also be used in the mechanical disturbance observation of other first-order systems.

TABLE II
PARAMETERS OF THE SPMSM OF THE PLATFORM

Parameters	Symbols	Value
Rated speed	n	3000[r/min]
Pole pairs	N_p	4
Rotational inertia	J	0.002[kg*m ²]
Stator resistance	R_s	0.91[Ohm]
Stator inductance	L_s	5.455[mH]
Permanent magnet flux	ψ_f	0.113[Wb]
Rated power	P_n	0.75[kW]
Rated load torque	T_L	[2Nm]
Viscous damping	B	0.001

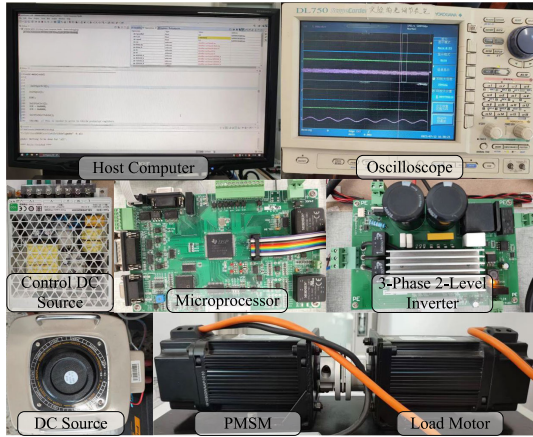


Fig. 8. Test bench of the SPMSM drive system.

V. EXPERIMENTAL EVALUATION

The proposed DSVM-based MPDSC and other comparative methods are investigated experimentally on a 0.75-kW SPMSM platform. The platform is shown in Fig. 8. The specifications of the SPMSM parameters are provided in Table II. The TMS320F28335 microprocessor from Texas Instruments (TI) performs the control strategies. The experimental results are recorded by the oscilloscope YOKOGAWA DL750 and transferred to the computer for better display and further analysis. The control frequency and sampling frequency used in the following algorithms are 10k if not otherwise specified. In the upcoming comparative performance experiments, to comprehensively assess the superiority of the proposed method across different performance metrics, only analogous methodologies from distinct sections will be used for a more meaningful comparison.

A. Steady-State Control Performance Comparison

In this section, the steady-state performance of the P-DSVM strategy is compared with that of C-MPCC and C-DSVM. For simplicity, “C-MPDSC” stands for the conventional MPDSC method [8], the MPDSC method proposed in [10] is referred to as “MPDSC-I,” the DSVM method proposed in [19] is used in the framework of MPDSC, which is referred to “MPDSC-II,” and “P-MPDSC” stands for the proposed method. The sampling time of the four control methods is 100 μ s. For a fairer comparison of the steady-state performance of the three above methods, the control frequency of the conventional MPDSC was set to 20k. For comparing the steady-state

performance of each control method more accurately, the pulsation amount of the dc variables is evaluated by the standard deviation, such as Δn , Δi_d , and Δi_q , which can be given as

$$\Delta X = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(x_i - \frac{1}{n} \sum_{i=1}^n x_i \right)^2} \quad (42)$$

where x_i is the i th value of the certain variable. The ac current is evaluated by the total harmonic distortion (THD). It is worth mentioning that in this article, the current and speed performance of the motor is explored as steady-state performance, and the thermal stability performance is not explored.

The steady-state performances are investigated under the three control methods at three operating conditions: 300-r/min speed with 0.5-N·m load, 1500-r/min speed with 1-N·m load, and 3000-r/min speed with 2-N·m load. Next, the waveforms of rotor speed, dq -axis currents, and A-phase current under different operating conditions are given in Figs. 9–11. Meanwhile, the standard deviation and THD of each variable for different operating conditions are shown in Table III. Since MPDSC-I, MPDSC-II, and P-MPDSC methods add the duty cycle or DSVM modulation method, their tracking performance is better than that of the C-MPDSC.

The experimental results of the speed of 300-r/min speed with 0.5-N·m load are shown in Fig. 9. The speed ripple in P-MPDSC decreased by 36.16% compared with MPDSC-I and decreased by 13.43% compared with MPDSC-II. The fluctuation in the d -axis current of P-MPDSC decreased by 38.41% compared with MPDSC-I and decreased by 8.07% compared with MPDSC-II. The fluctuation in the q -axis current of P-MPDSC decreased by 33.44% compared with MPDSC-I and decreased by 11.7% compared with MPDSC-II. The THD of the phase current in P-MPDSC decreased by 34.42% compared with MPDSC-I and by 2.85% compared with MPDSC-II.

Similarly, the conditions of the speed of 1500-r/min speed with 1-N·m load and the speed of 3000-r/min speed with 2-N·m load are shown in Figs. 10 and 11, respectively. In Fig. 10, the speed fluctuation of P-MPDSC reduced by 27.09% compared with MPDSC-I and decreased by 1.69% compared with MPDSC-II. The THD of phase current in P-MPDSC decreased by 6.02% compared with MPDSC-I and by 1.06% compared with MPDSC-II. In Fig. 11, the speed fluctuation of P-MPDSC reduced by 35.12% compared with MPDSC-I and decreased by 6.19% compared with MPDSC-II. The THD of phase current in P-MPDSC decreased by 6.44% compared with MPDSC-I and decreased by 1.52% compared with MPDSC-II.

It also has a more precise speed and current tracking performance in this condition for the P-MPDSC. Above all, the steady-state control performance of the proposed MPDSC method has better steady-state performance.

B. Computational Burden Comparison

The computational burden of the algorithm implementation is a significant factor limiting the application of MPC

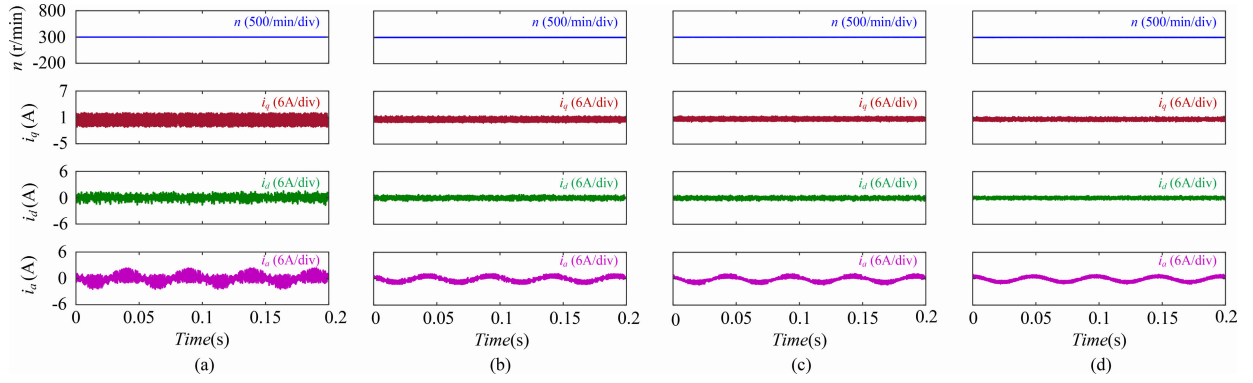


Fig. 9. Experimental result of the steady-state response at 300-r/min speed with 0.5-N·m load. (a) C-MPDSC, (b) MPDSC-I, (c) MPDSC-II, and (d) P-MPDSC.

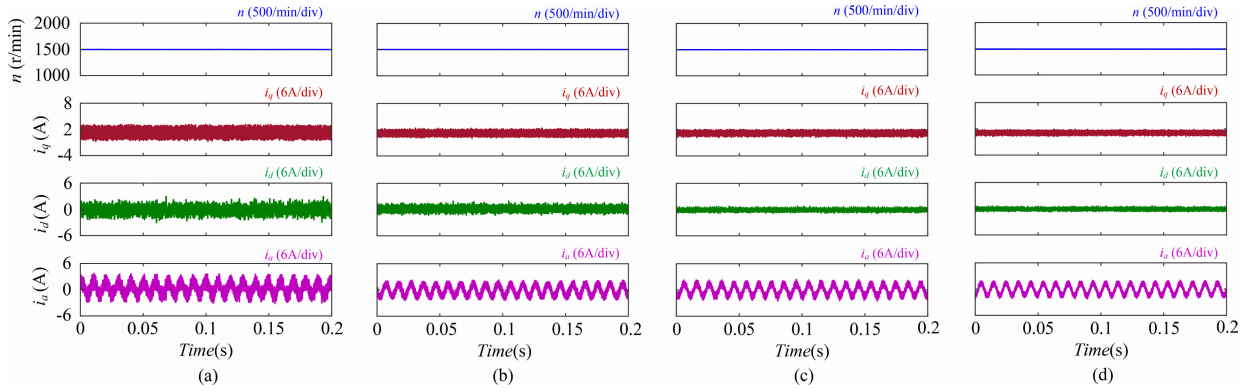


Fig. 10. Experimental result of the steady-state response at 1500-r/min speed with 1-N·m load. (a) C-MPDSC, (b) MPDSC-I, (c) MPDSC-II, and (d) P-MPDSC.

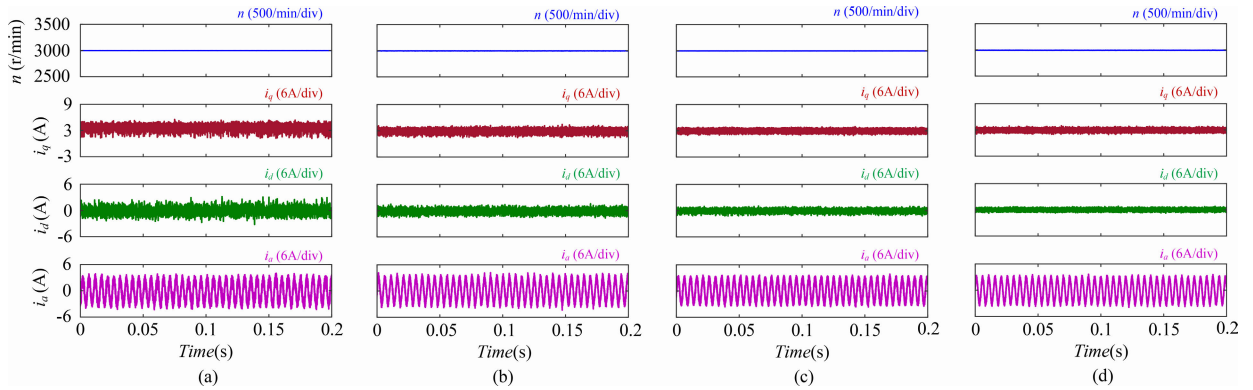


Fig. 11. Experimental result of the steady-state response at 3000-r/min speed with 2-N·m load. (a) C-MPDSC, (b) MPDSC-I, (c) MPDSC-II, and (d) P-MPDSC.

methods in complex systems. The increase in computing time means that more expensive hardware is needed. To validate the simplicity of the proposed DSVM method, this section compares the method proposed in this article with other DSVM methods [15], [19] and the conventional MPDSC [8]. The DSVM method mentioned in [15] is denoted as DSVM-I, and the DSVM method proposed in [19] is one of the three methods shown in Fig. 12. The input/output (I/O) port of the microprocessor is used to evaluate the execution time. After the algorithm is running, set the level of the GPIO port to high, reset it to low after the test algorithm is complete, and then use an oscilloscope to measure the width of the high level to measure the execution time. To fairly measure the execution time of each different method, the execution time is

the average of multiple experiments under various operating conditions for each method.

For all four methods, the sampling process and coordinate transformations were set to be the same, with a common total time of 18.7 μ s. This is because the experimental validation process for each method uses the same software and hardware. The execution time of C-MPDSC is 23.0 μ s, which is lower than that of all other DSVM-based methods. This is because C-MPDSC only needs to evaluate seven VVs and possesses poor steady-state performance. Combined with the steady-state performance results in Section V-A, the proposed method increases 3.8 μ s over C-MPDSC, but the steady-state performance is effectively improved. Among the DSVM methods, the algorithm execution time of the P-MPDSC

TABLE III
EXPERIMENTAL VALUES OF STEADY-STATE PERFORMANCE

Conditions	Indexes	C-MPDSC	MPDSC-I	MPDSC-II	P-MPDSC
300 rpm, 0.5 Nm	$\Delta n(\text{r/min})$	0.7205	0.3443	0.2539	0.2198
	$\Delta i_d(\text{A})$	0.6880	0.2997	0.2008	0.1846
	$\Delta i_q(\text{A})$	0.6381	0.2823	0.2130	0.1879
	THD(%)	40.96	14.06	9.49	9.22
1500 rpm, 1 Nm	$\Delta n(\text{r/min})$	0.6165	0.1912	0.1418	0.1394
	$\Delta i_d(\text{A})$	0.4938	0.1723	0.1431	0.1402
	$\Delta i_q(\text{A})$	0.5812	0.1767	0.1359	0.1353
	THD(%)	24.76	11.97	11.37	11.25
3000 rpm, 2 Nm	$\Delta n(\text{r/min})$	0.6404	0.2244	0.1552	0.1456
	$\Delta i_d(\text{A})$	0.6126	0.1859	0.1581	0.1558
	$\Delta i_q(\text{A})$	0.5731	0.3107	0.2364	0.1987
	THD(%)	19.80	9.01	8.56	8.43

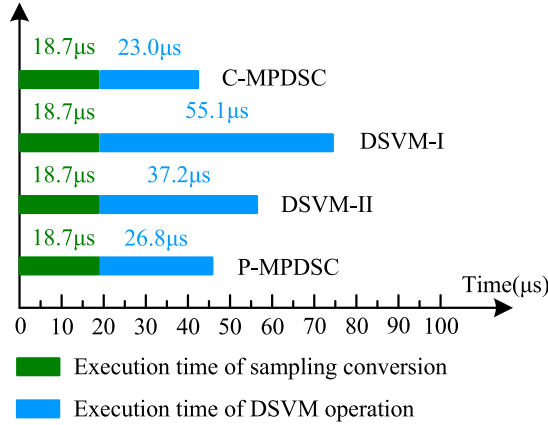


Fig. 12. Experimental result of the execution time of DSVM-I, DSVM-II, and P-MPDSC.

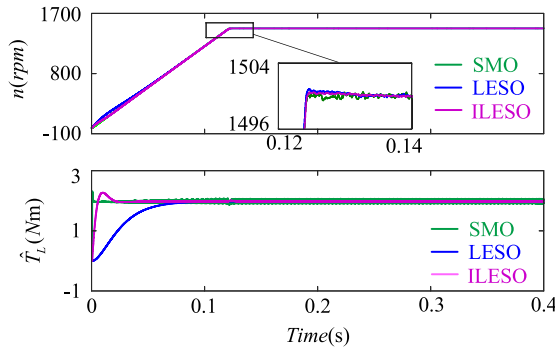


Fig. 13. Experimental result for the start performance of three load observers between the SMO-based, the LESO-based, and the ILESO-based load observer from stop to the 1500-r/min speed and 2-N-m load.

mentioned is $28.3 \mu\text{s}$ less than that of DSVM-I and $10.4 \mu\text{s}$ less than that of DSVM-II. This execution time reduction is due to the optimal VV determination process being optimized by the three auxiliary lines without any LUTs and cascaded cost functions. From Section IV and this section, the proposed P-MPDSC can achieve more satisfactory steady-state performance with lower computational effort.

C. Dynamic Responses Performance Under Different Conditions

To verify the dynamic response of the proposed DSVM and the improved load disturbance observer in this article,

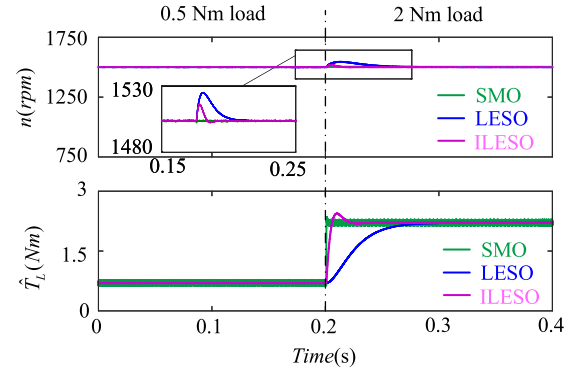


Fig. 14. Experimental result for the sudden load performance of three load observers between the SMO-based, the LESO-based and the ILESO-based load observer from the 0.5- to 2-N-m load.

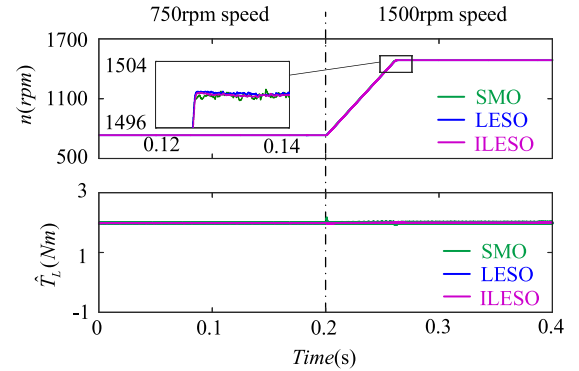


Fig. 15. Experimental result for the sudden speed performance of three load observers between the SMO-based, LESO-based, and ILESO-based load observer from 750 to 1500 r/min.

the SMO-based load observer proposed in [24] and the conventional LESO-based load observer proposed in [26] are used to compare the related performance. The SMO-based load observer is denoted as SMO, the conventional LESO-based load observer is denoted as LESO, and the load observer proposed in this article is denoted as ILESO. The gain of the SMO is selected as 10, and the gain factor ω_0 of the conventional ESO-based load disturbance observer and the improved one is set as the same 30. However, in real motor systems, it is necessary to choose the appropriate bandwidth based on the environmental noise level to compromise the fast performance of the observer and the effect of noise. It is worth

mentioning that the above-mentioned observer has no effect on the efficiency of the motor system.

The start performance of 1500-r/min speed and 2-N·m load of the proposed DSVM method with the three above load disturbance observers is shown in Fig. 13. From Fig. 13, the speed performance of SMO and ILESO is quickly converging approximately, and the LESO has a slower dynamic convergence process than the other two methods. SMO-based load observer can achieve a quicker convergence process, but there is an unacceptable fluctuation worse than LESO and ILESO when it is stable. Although LESO is able to estimate the load torque decently when it is stabilized, its dynamic convergence process remains slow compared with ILESO. In contrast, ILESO not only guarantees good estimation accuracy but also has a faster convergence process. The performances of sudden load from 0.5 to 2 N·m between the three observers are shown in Fig. 14. The performances of sudden speed from 750 to 1500 r/min between the three observers are shown in Fig. 15. From Fig. 14, under the sudden load, the conventional LESO-based load observer will have a large overshoot in speed performance, while ILESO will be relatively small. From Fig. 15, under sudden speed, the speed performances of SMO and ILESO are similar, and the LESO is a more slowly dynamic convergence process than the other two methods. When estimating the load, similar to the conclusions obtained in Fig. 13, both Figs. 14 and 15 show that ILESO-based load observer achieves better estimation than SMO while achieving a faster dynamic convergence process with a similar estimate precision as LESO. In summary, the proposed observer may effectively improve the dynamic performance of speed control and mechanical disturbance estimation performance.

D. Robustness of ESOs to Variations in the Moment of Inertia J

To evaluate the disturbance rejection performance of the conventional ESO-based load observer and the load observer proposed in this article, this part investigates the effect of the mismatch of the inertia parameter J on the performance of the above two observers. The inertia values used in the control system are set to 0.8 times the nominal value ($0.8 J_0$), the nominal value (J_0), and 1.2 times the nominal value ($1.2 J_0$) in three cases. The working condition of the motor was set to load 2 N·m and accelerate from 750 to 1500 r/min in 0.05 s.

Fig. 15 shows the transient performance for different values of inertia J_0 . The experimental results of the IESO show in Fig. 16(a), and the experimental results of the conventional ESO show in Fig. 16(b). The load estimation produces a fixed error during the acceleration process due to the fixed variation in the inertia J . A fixed multiplicity of J_0 causes the mechanical equations to no longer hold in practice, but the ESOs will estimate it as an extra mechanical disturbance and estimate this error into the mechanical lump disturbance. Comparing the experimental speed response result of the conventional ESO-based load observer and the proposed one, the proposed load observer in this article can maintain a faster convergence speed in the case of inertia mismatch. Therefore, it can be concluded that both the observers are robust to the

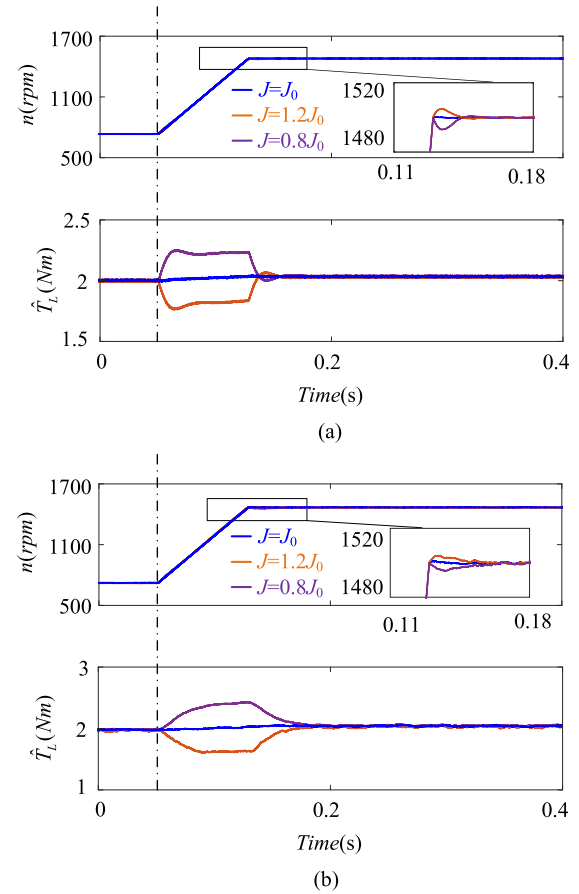
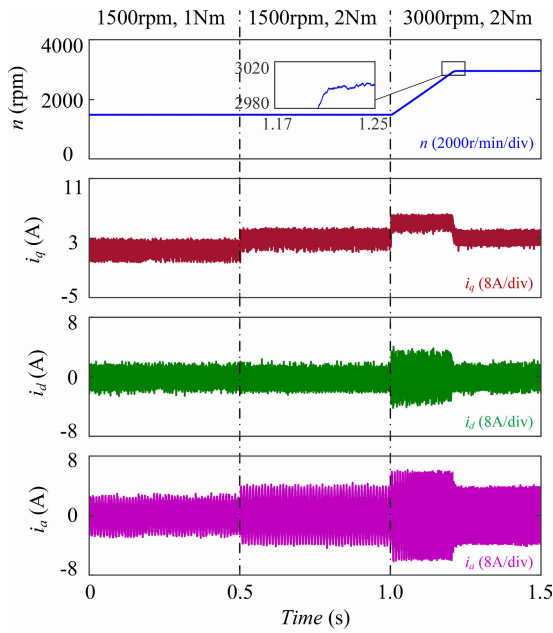


Fig. 16. Transient performance of step change in speed with different values of inertia. (a) IESO. (b) Conventional ESO.

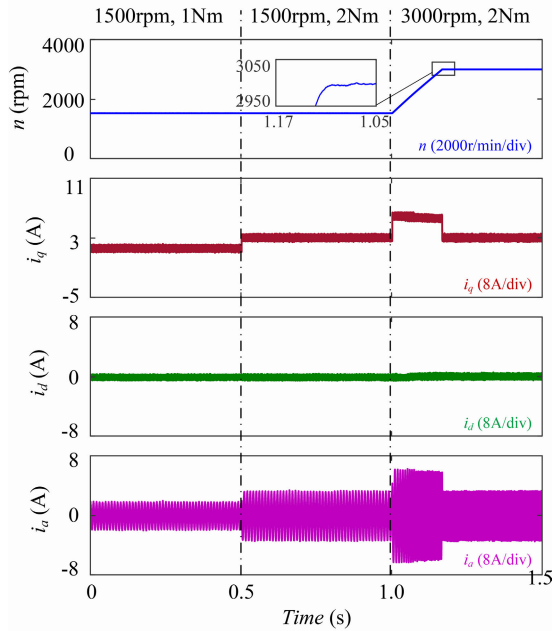
mismatch of rotational inertia J , while the proposed observer has better fast response capability.

E. Dynamic Responses Performance Between C-MPDSC and Proposed Method

In addition to comparing the proposed DSVM method performance and observer performance, this section also tests the dynamic control performance of C-MPDSC and P-MPDSC under sudden load and acceleration conditions. Fig. 17(a) and (b) depicts the dynamic response of C-MPDSC and P-MPDSC with the condition of 1500 r/min 1 N·m to 1500 r/min 2 N·m and the condition of 1500 r/min 2 N·m to 3000 r/min 2 N·m. The proposed method is similar to the traditional method in the convergence process of current and speed during the load change process from 1 to 2 N·m. The dynamic performance of current and speed of the P-MPDSC during the variable load from 1 to 2 N·m is similar to C-MPDSC. However, in the process of accelerating from 1500 to 3000 r/min, the current distortion of the C-MPDSC is obvious, and the d -axis fluctuation is obviously increased, while the P-MPDSC can still maintain good performance. Then, according to the above experimental results, the proposed MPDSC method can achieve the better steady-state performance under similar dynamic control performance.



(a)



(b)

Fig. 17. Experimental results of the dynamic performance (a) C-MPDSC and (b) P-MPDSC.

F. Disturbance Rejection Evaluation of the Load Disturbance Observer

To further explore the disturbance rejection ability of the traditional ESO-based load observer and the ESO-based load observer proposed in this article, this section explores and evaluates the effects of speed measurement disturbances on both the observers. This experiment is designed to simulate a speed measurement disturbance by adding a random number of $[-2, 2]$ to the speed sampling value at 0.2 s under a motor operating condition of 1500 r/min 2 N·m and to monitor the load observations.

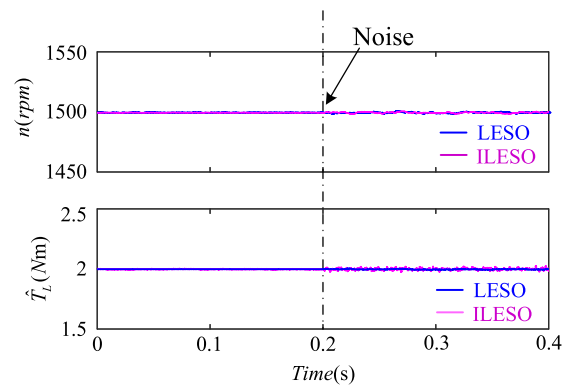


Fig. 18. Experimental results of the conventional ESO-based observer and the IESO-based observer with the measurement noise.

The obtained experimental results are shown in Fig. 18, which shows the load estimations of the conventional ESO-based load observation and the proposed load observer after using the speed sampled values with added noise. However, due to its increased bandwidth, the proposed load observer is more sensitive to high-frequency noise at speed. This phenomenon is consistent with the frequency-domain analysis in Section III regarding the two observers. Nevertheless, according to the previous experiments related to the speed response, the load observer proposed can have a faster response, which is more catering to the fast characteristics of MPDSC.

VI. CONCLUSION

A modified DSVM-based MPDSC with improved load disturbance observer was proposed for PMSM drive systems to achieve a more precise tracking performance and high mechanical disturbance estimation performance. First, a simplified DSVM method was proposed based on the three auxiliary lines in different sectors. In this way, the complexity and the computational burden of DSVM were reduced. Then, based on the estimated error of the conventional LESO-based load disturbance observer, an improved load disturbance observer is proposed to accelerate the convergence of the observer. The experimental results showed that the proposed method has better tracking performance, lower computational burden, and excellent mechanical disturbance estimation ability than the conventional method.

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